

Suman Chowdhury



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github/scinv22



linkedin/scinv22



scinv22.github.io

RESEARCH FOCUS

EVENT RECOGNITION AND REASONING • CAUSAL REASONING • SCENE UNDERSTANDING • AUTONOMOUS AGENTS

EDUCATION

PH.D. IN COMPUTER SCIENCE (COTUTELLE)

July 2023 – Present

ROYAL MELBOURNE INSTITUTE OF TECHNOLOGY (RMIT UNIVERSITY)

ACADEMY OF SCIENTIFIC AND INNOVATIVE RESEARCH (ACSIR)

- Advisors: Dr. Rajesh P. Barnwal (AcSIR), Dr. Haytham Fayek (RMIT), and Prof. Feng Xia (RMIT)

- Cumulative GPA: 8.57/10.0

- Ph.D. Dissertation (Tentative): Causal Scene Understanding in Autonomous Agents

M.TECH IN MECHATRONICS

Aug 2021 – June 2023

ACADEMY OF SCIENTIFIC AND INNOVATIVE RESEARCH (ACSIR)

- Advisor: Dr. Rajesh P. Barnwal (AcSIR)

- Cumulative GPA: 8.44/10.0

- M.Tech Thesis: Multi-Label Classification for Road Scene Understanding

B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING

Aug 2015 – Jun 2019

JALPAIGURI GOVERNMENT ENGINEERING COLLEGE, WB

- Cumulative GPA: 7.99/10.0

- B.Tech Project: Distributed Storage System using Optimal Exact-Regenerating Codes

RESEARCH EXPERIENCE

CSIR SENIOR RESEARCH FELLOW (GATE)

Nov 2023 – Present

AI & IoT LAB, IT GROUP, CSIR-CMERI, DURGAPUR, INDIA

- Leading research on traffic scene understanding, event recognition, and causal reasoning for autonomous vehicles.

- Current focus: integrating Vision-Language Models (VLMs) and Scene Graphs for interpretable AV decision-making.

- Completed first PhD objective on multi-label activity recognition in traffic scenes (under review at IEEE TCSVT).

CSIR JUNIOR RESEARCH FELLOW (GATE)

Nov 2021 – Nov 2023

AI & IoT LAB, IT GROUP, CSIR-CMERI, DURGAPUR, INDIA

- Developed multi-label learning framework for road scene understanding.

- Initiated survey on high-level scene understanding in traffic environments (under review at ACM CSUR).

SUMMER RESEARCH INTERN (UG)

Jun 2018

DEPT. OF EECE, IIT KHARAGPUR, INDIA

- Built a practical distributed storage system prototype using exact-regenerating codes.

- Implemented data storage/retrieval and node failure recovery.

- Advisor: Prof. Sant Sharan Pathak

PUBLICATIONS

ACCEPTED MANUSCRIPTS

Sahu, Bidya, Suman Chowdhury, and Rajesh P. Barnwal. "Near Real Time Detection of APT Attack in Industrial IoT Environment."

Conference: ICDCN 2026

Chowdhury, Suman, Farook, Yahya, and Rajesh P. Barnwal. "Real-Time Vision-based Scene Classification for Adaptive Behavior in Autonomous Mobile Robots."

Conference: ACM CODS-COMAD December-2024

Chakraborty, Koustav, Suman Chowdhury, and Rajesh P. Barnwal. "Study on the Efficacy of GPT Double Heads Model for Question-Answering in Bengali."

Conference: IEEE Calcon 2024

MANUSCRIPTS UNDER REVIEW

Chowdhury, Suman, Apoorv Sisodia, Prerana Mukerjee, and Rajesh P. Barnwal. "IDD-ER: A Diagnostic Dataset for Relational Event Reasoning in Unstructured Driving Scenes for Autonomous Agents."

IEEE Robotics and Automation Letters (IEEE R-AL)

Chowdhury, Suman, Prerana Mukerjee, and Rajesh P. Barnwal. "GraSS: Graph-Guided Semantic Slot Attention for Efficient Multi-Label Activity Recognition in Autonomous Driving."

IEEE Transactions on Intelligent Transportation Systems (IEEE T-ITS)

Chowdhury, Suman, Prerana Mukerjee, and Rajesh P. Barnwal. "Vision-Based High-Level Road Scene Understanding for Autonomous Vehicles: State of the Art, Challenges, and Future Directions."

ACM Computing Surveys (ACM CSUR)

Chowdhury, Suman, and Rajesh P. Barnwal. "Empirical Study of Transformer and Graph-Based Architectures for Class-Imbalanced Road Scene Recognition." Available at SSRN 4718551.

MENTORING EXPERIENCE

Supervised multiple B.Tech interns (2023–25) on projects in deep learning, scene understanding, and vision-based AVs.

Outcomes include 10 completed projects and 3 accepted conference publications.

ACHIEVEMENTS

Awarded Travel Grant – ACM CODS-COMAD 2024 Conference

December 2024

Selected for a travel grant to attend the ACM CODS-COMAD conference.

- Link: cods-comad.in/dec-2024/travel-grants.php

ACM India Anveshan Setu Fellowship 2024–25

February 2024

Selected among 46 CS/IT PhD scholars across India for a prestigious mentorship program with leading researchers. As part of the fellowship, visited Prof. Prerana Mukherjee at Jawaharlal Nehru University in March 2025.

- Link: acm.org/articles/acm-india-bulletins/2024/announcing-acm-india-anveshan-setu-fellows

CSIR Junior Research Fellowship (GATE)

November 2021 – November 2026

Awarded to a maximum of five scholars per CSIR national laboratory to pursue the Integrated M.Tech–PhD program at AcSIR through GATE qualification.

Full Fee Scholarship (FFS)

August 2015 – June 2019

Granted full tuition fee waiver for the 4-year B.Tech program at Jalpaiguri Government Engineering College by the Government of West Bengal, based on academic merit and financial need.

JEE Advanced 2015 – Common Merit List

June 2015

Secured AIR 16,887 (General category) out of 1.2 million candidates. Also qualified the Architecture Aptitude Test (AAT) 2015, organized by IIT Bombay.

SKILLS

PROGRAMMING

Experienced: Python | GNU Octave • Familiar: C | C++ | MATLAB | JAVA

FRAMEWORKS & LIBRARIES

PyTorch • Tensorflow • Numpy • Pandas • Scikit-learn • Matplotlib

LANGUAGES

Native: Bengali • Fluent: English | Hindi

VOLUNTEERING

Secretary

JULY 2024 - PRESENT

AcSIR Science Club

CSIR-Central Mechanical Engineering Research Institute

Teacher

JANUARY 2016 - JUNE 2019

Jyoti- A free Night School for underprivileged students

Jalpaiguri Government Engineering College